

Indian Institute of Engineering Science and Technology, Shibpur
CST Third Semester Examination, November 2024
Signals and Systems (CS-2104)

Plan-A: Answer any 5 out of 8 questions, $5 + 5 = 10$ marks.

Plan-B: Answer any 10 subparts out of the 16 subparts of 8 Qs.

Full Marks 50

Time 180 minutes

1. (a) Derive the Fourier transform of $\int f(t)dt$ and $\frac{d}{dt}f(t)$ from basic definitions of Fourier transform and inverse Fourier transform in continuous domain.

(b) Explain the duality relation between rectangular unit pulse for $|t| < T_1$ and bandlimited rectangular frequency spectrum with unit magnitude within bandwidth $|\omega| < W$.

2. (a) Find the Fourier components of the discrete signal:

$$x[n] = 1 + \sin\left(\frac{2\pi}{N}\right)n + 3\cos\left(\frac{2\pi}{N}\right)n + \cos\left(\frac{4\pi}{N} + \frac{\pi}{2}\right)n.$$

Hint: This signal is periodic in N and may be expanded into complex exponentials for simplification.

(b) Perform convolution for an LTI system $x[n] = 2^n u[-n]$ with unit impulse response $h[n] = u[n]$.

Hint: Check the convolution output $y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$ in the intervals $n \geq 0$ and $n < 0$.

3. (a) Compare the performance of amplitude modulation system with that of frequency modulation system.

(b) Describe various frequency demodulation techniques and explain how additive noise is tackled.

4. (a) What is the impulse response of a system described by first order difference equation $y[n] - ay[n-1] = x[n]$? Explain the design of low pass filter and high pass filter, considering an input signal $x[n] = e^{j\omega n}$.

(b) Find the transfer function and solution for the discrete system described by $x(k+1) = Fx(k) + Hu(k)$ and $y(k) = Cx(k)$ where $x(k)$ is $n \times 1$ state vector, $u(k)$ is $m \times 1$ input vector, $y(k)$ is $p \times 1$ output vector, and F, H, C being matrices of appropriate dimensions.

5. (a) For a first order system $\tau \frac{dy(t)}{dt} + y(t) = x(t)$, find expressions for the magnitude and phase of the frequency response and provide rough sketch indicating the filtering properties.

(b) Obtain the state space equations and hence the state transition matrix for this second order system described by the differential equation:

$$\ddot{x} + 10\dot{x} + 21x = y \text{ i.e. } \frac{d^2x}{dt^2} + 10\frac{dx}{dt} + 21x = y$$

6. (a) Describe how multiplexing of signals is achieved in time domain and frequency domain.

(b) Describe the technique of pulse amplitude modulation and its utility in communication system.

7. Explain the phenomenon in mathematical terms:

(a) Positive feedback due to close proximity of speaker and microphone

(b) Stability analysis of discrete dynamical systems

8. Write short notes on **maximum two** of the following:

(a) Numerical computer algorithms for state feedback

(b) Amplitude modulation and demodulation systems

(c) Linear time invariant systems and their properties